Permutation Tests: Through Part D

Mark M. Fredrickson (mfredric@umich.edu)

Module 3: Resample, Session 4

Computational Methods in Statistics and Data Science (DATASCI 406)

Today

No video before this session. Each idea here is followed by the Part D item that uses it.

Idea	Then
flip signs under symmetry	D.1(a)
invert the test	D.1(b)
shuffle labels under $F = G$, choose T	D.2(a)
ranks	D.2(b)
a parameter null	D.3

The full deck is the video, posted before Tuesday. Turn in `project_d.html` on Canvas.

The recipe: condition, rearrange, compute, count

Four things to name, every time:

1. H_0: some rearrangement leaves the data's distribution unchanged.
2. **Condition on** what H_0 says nothing about.
3. **Rearrange** the rest. Given step 2, every rearrangement is equally likely under H_0.
4. T: compute it on every rearrangement, or on B random ones.

p-value: the fraction of rearrangements with T at least the observed value.

No F is needed, and the test is exact.

H_0 picks the rearrangement, the alternative picks T

- The **rearrangement** comes from H_0. It makes the level of the test exact.
- The **statistic** comes from the alternative. It decides which departures from H_0 the test can see.

Today: **flip signs** and **shuffle labels**.

Rearrangement 1: Signs

Under symmetry, the signs are fair coins given the magnitudes

$D_1, \ldots, D_n$ independent, not necessarily identically distributed.

$$H_0 : D_i - \theta_0 \text{ and } -(D_i - \theta_0) \text{ have the same distribution, for each } i.$$

Write $D_i - \theta_0 = S_i \, |D_i - \theta_0|$, $S_i = \pm 1$. Under H_0, given the magnitudes, the S_i are independent fair coins.

- **Condition on**: the magnitudes $|D_i - \theta_0|$.
- **Rearrange**: the signs. All 2^n patterns equally likely.
- T: $\frac{1}{n} \sum_i (D_i - \theta_0)$, the mean of the signed differences.

If flushing does nothing, D_i is symmetric about 0

Deck 1: $D_i = \log X_i - \log Y_i$, first draw against after 2 minutes of flushing, in 271 homes.

H_0: flushing makes no difference, so the two bottles from a home are **exchangeable**: (X_i, Y_i) and (Y_i, X_i) have the same distribution.

Then $\log X_i - \log Y_i$ and $\log Y_i - \log X_i$ have the same distribution: D_i is symmetric about $\theta_0 = 0$.

```
d <- log(vt$first_draw) - log(vt$flush_2min)
round(d[11:20], 2)
```

```
##  [1]  3.39  1.72  1.53  1.32  1.48 -0.15  1.75  0.42  1.52
## [10]  1.40
```

Ten homes: all 1024 patterns, $p = 2/1024$

`expand.grid` lists every combination of its arguments, so each row of `S` is one of the $2^{10} = 1024$ sign patterns.

```
d10 <- d[11:20]
S <- as.matrix(expand.grid(rep(list(c(-1, 1)), 10)))
null10 <- as.vector(S %*% abs(d10)) / 10
c(observed = mean(d10), p = mean(null10 >= mean(d10) - 1e-12))
```

```
## observed         p
## 1.437092 0.001953
```

One of the ten is negative. Only the observed pattern and the all-positive one give T that large.

271 homes: draw patterns at random

2^{271} patterns are too many to list. Draw B at random, and use the $(1 + \#)/(B + 1)$ p-value from module 1.

```
t_obs <- mean(d)
t_flip <- replicate(B, mean(sample(c(-1, 1), n, replace = TRUE) * abs(d)))
c(observed = t_obs, max_null = max(t_flip),
  p = (1 + sum(t_flip >= t_obs)) / (B + 1))
```

```
##  observed  max_null         p
## 1.4278869 0.3384585 0.0002499
```

The null is symmetric about 0, so a two-sided test compares $|T^*|$ with $|T|$.

Now: D.1(a)

Saturday minus the Sunday after it, 105 weekends. H_0: each D_i symmetric about 0.

- `d <- bikes$total[sat] - bikes$total[sat + 1]`
- $B = 10{,}000$ random sign patterns on `abs(d)`, $T = \bar{D}^*$.
- Two-sided: $(1 + \#\{|T^*| \geq |T|\})/(B + 1)$.

Name the four cells before you run it. Not `sample(d)`: that keeps every value and gives $p = 1$.

Check: $T \approx 320$, $p \approx 0.009$.

Inverting the test gives an interval

Keep every θ_0 the two-sided 5% test does not reject. The kept values are a 95% interval.

At θ_0, flip the signs of $d_i - \theta_0$. One sign matrix serves every θ_0:

$$T_b^*(\theta_0) = \frac{1}{n}\sum_i S_{bi}(d_i - \theta_0) = \underbrace{\frac{1}{n}\sum_i S_{bi} d_i}_{\texttt{S \%*\% d / n}} - \theta_0 \ \underbrace{\frac{1}{n}\sum_i S_{bi}}_{\texttt{rowMeans(S)}} .$$

The null is symmetric about 0, so compare $|T^*|$ with $|T|$.

One sign matrix serves every θ_0

```
S <- matrix(sample(c(-1, 1), B * n, replace = TRUE), B)
Sd <- as.vector(S %*% d) / n
Sbar <- rowMeans(S)
p_two <- function(theta0) {
  t0 <- mean(d) - theta0
  (1 + sum(abs(Sd - theta0 * Sbar) >= abs(t0))) / (B + 1)
}
grid <- seq(1.2, 1.7, by = 0.002)
range(grid[map_dbl(grid, p_two) > 0.05])
```

```
## [1] 1.290 1.564
```

Now: D.1(b)

- `S`: 10,000 $\times$ 105 random signs. `Sd <- S %*% d / 105`, `Sbar <- rowMeans(S)`.
- `p_two(theta0)`: two-sided p-value of $T = \bar{d} - \theta_0$.
- `map_dbl` over `seq(-200, 800, by = 2)`. Keep $p > 0.05$, report the range.
- Compare with `t.test(d)$conf.int`.

Check: the two intervals nearly agree, about 80 to 565 rentals.

Rearrangement 2: Labels

Under $F = G$, every split of the pooled values is equally likely

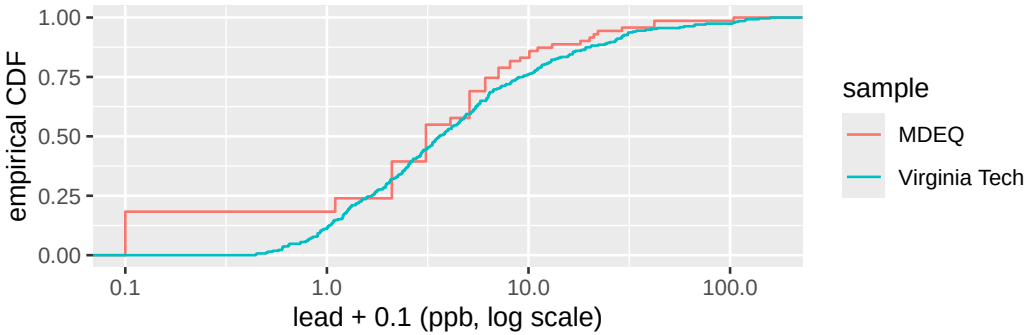
$X_1, \ldots, X_m \overset{\text{iid}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} G$, independent.

- H_0: $F = G$. The $N = m + n$ pooled values are then iid, and the labels carry no information.
- **Condition on**: the pooled values.
- **Rearrange**: the labels. Each of the $\binom{N}{m}$ splits into m and n is equally likely.
- T: $\bar{X} - \bar{Y}$, to start.

Shuffle the labels, recompute T, count.

Did MDEQ and Virginia Tech sample the same distribution?

In 2015 MDEQ collected 71 samples, and Virginia Tech collected 271 first-draw samples.



Shuffle the labels: the difference in means does not reject

g is TRUE for the MDEQ samples. sample(g) shuffles the labels and keeps the group sizes.

```
diff_means <- function(z, g) mean(z[g]) - mean(z[!g])
t_obs <- diff_means(z, g)
t_perm <- replicate(B, diff_means(z, sample(g)))
c(observed = t_obs, p = (1 + sum(abs(t_perm) >= abs(t_obs))) / (B + 1))
```

```
## observed        p
##  -3.3361   0.2189
```

The rearrangement gives the level, the statistic gives the power

H_0 fixes the rearrangement. Any T gives an exact test from the same shuffles.

- **Difference in means**: a shift in the center. Sensitive to the largest values.
- **Difference in medians**: a shift in the middle.
- **Kolmogorov-Smirnov**, $D = \max_x |\hat{F}_m(x) - \hat{G}_n(x)|$: any difference between the two CDFs.

Choose T before looking. Trying several and reporting the smallest p-value invalidates the test.

Three statistics from the same shuffles

```
ks_stat <- function(z, g) max(abs(ecdf(z[g])(z) - ecdf(z[!g])(z)))
three <- function(z, g) c(mean = diff_means(z, g),
                          median = median(z[g]) - median(z[!g]),
                          ks = ks_stat(z, g))
t3 <- three(z, g)
perm3 <- replicate(B, three(z, sample(g)))
(1 + rowSums(abs(perm3) >= abs(t3))) / (B + 1)
```

```
##    mean  median      ks
## 0.23319 0.53812 0.03999
```

KS finds the gap at 0, a difference in recording

```
v <- sort(unique(z))
gap <- ecdf(z[g])(v) - ecdf(z[!g])(v)
c(at = v[which.max(abs(gap))], gap = gap[which.max(abs(gap))])
```

```
##     at    gap
## 0.0000 0.1831
```

- MDEQ recorded whole numbers, 13 of them 0. No Virginia Tech sample is below 0.344 ppb.
- The null is **equal distributions**. A T that sees the whole distribution finds any difference, including one in recording.

Now: D.2(a)

Working days against weekends and holidays, `total` rentals. H_0: $F = G$.

- `three(z, g)`: difference in means, difference in medians, KS at `sort(unique(z))`.
- $B = 4000$ shuffles of g, all three statistics on each, three p-values.
- Where are the two CDFs farthest apart, and what fraction of each group is below it?

Check: only KS rejects, p near 0.10, 0.46 and 0.04. The gap is near 3500 rentals, 27% of working days at or below it against 38% of the others.

Ranks: same H_0, same shuffle, a T that forgets the data

- Replace each value by its **rank** in the pooled sample, from 1 to N. $T = W$, the sum of the X ranks.
- Under H_0, with no ties, the X ranks are a random m-subset of $\{1, \ldots, N\}$.
- The null depends only on m and n, so it could be tabulated. That is what rank tests were for.

```
r <- rank(z); w_obs <- sum(r[g]); mid <- sum(g) * (length(z) + 1) / 2
w_perm <- replicate(B, sum(r[sample(g)]))
c(permutation = (1 + sum(abs(w_perm - mid) >= abs(w_obs - mid))) / (B + 1),
  wilcox = wilcox.test(mdeq$lead, vt$first_draw)$p.value)
```

```
## permutation      wilcox
##      0.2272      0.2241
```

Now: D.2(b)

- `rank(z)`, and W = the sum of the working-day ranks.
- Center at $m(N+1)/2$, shuffle $B = 4000$ times, two-sided p-value on $|W^* - m(N+1)/2|$.
- Compare with `wilcox.test(z[g], z[!g])$p.value`.

Check: both p-values near 0.12, and neither rejects.

Equality or a Parameter?

$F = G$ is stronger than $\theta_F = \theta_G$

- The shuffle tests **equality**, $F = G$, exactly.
- The question is often a **parameter**: $\theta_F = \theta_G$, the two means, say.
- $F = G$ implies $\theta_F = \theta_G$, not the reverse.
- Under $\theta_F = \theta_G$ with $F \neq G$, shuffled data do not look like the original. The level is no longer exact.

Equal means, unequal spreads

- Shuffling pools the variances: the permutation SE of $\bar{X} - \bar{Y}$ uses $\hat{\sigma}^2_{\text{pooled}}(1/m + 1/n)$. The truth is $\sigma_X^2/m + \sigma_Y^2/n$.
- **Small group, large spread**: the permutation SE is too small and the test over-rejects. The other way, it under-rejects.
- **Welch's** t divides by $\sqrt{s_X^2/m + s_Y^2/n}$, each group's own variance. Its true and permutation nulls both approach $N(0,1)$, so the level holds in the limit.
- For $F = G$, any T. For a parameter, studentize.

Now: D.3

Casual riders differ between the groups in mean **and** spread. Build populations with equal means and the observed spreads, and check the level.

- Center `casual` within each group. Report the sizes and SDs.
- $R = 500$ datasets: resample each group's centered values with replacement.
- In each, a permutation test with $B = 400$ shuffles for the difference in means and for Welch's t. Record $p \leq 0.05$.
- Report the two rejection rates. Which group is small with a large spread?

Check: the 231 other days have SD about 870, the 500 working days about 390. The difference in means rejects about 14% of the time, Welch's about 6%.

Summary: the four cells

H_0	condition on	rearrange	T
D_i symmetric about θ_0	the magnitudes	the signs, 2^n patterns	$\bar{D}$
$F = G$	the pooled values	the labels, $\binom{N}{m}$ splits	any: mean, median, KS, rank sum
$\theta_F = \theta_G$ only	the pooled values	the labels, no longer equally likely	studentize

- The rearrangement comes from H_0 and sets the level.
- The statistic comes from the alternative and sets the power.
- Invert the test for an interval. One matrix serves every θ_0.